

IT5106 - Software Development Project

Project Proposal - 2025

Candidate Details

Index No: 2226804

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Supervisor/Advisor Details

	Supervisor 1 (IT Related)	Supervisor 2 (Optional)
Name	JAM. Sanjeewa	
Designation	Lecturer	
Workplace address	No 260, Dr Danister De Silva Mawatha, Colombo 09.	
Academic/ professional qualifications and memberships	-	

Work experience	5+ years' experience in implementing IT project	
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Note: Any deviation of the final project from the project described in this proposal should be explained by the candidate in the final Project Report.

Project Details

1. Title of Project

Art Gallery Management System for “Galaxy of Art” Shop

2. Client Information (If applicable)

- **Name:**
- **Address:**
- **Contact Information:**

3. Introduction

Galaxy of Art is a popular Art Gallery shop which has many kind of hand paintings, mugs printings and some kind of statues. Currently they are not using computerized system. All tasks are done by humans. My new proposed system designed to streamline, automate and enhance the overall daily operations like selling, production and purchasing experience for both members and customers.

The system supports the management of hand paintings, statues and mug printing services with business functions such as supplier management, production management, inventory control, quotation handling, payment tracking, invoicing and report generations and so on.

All information are stored within a secure database. And this system aims to replace manual record-keeping with a centralized platform that ensure accuracy, efficiency and real-time data accessibility. This enables to continuously maintaining a clear business processes along with data management and analysis.

4. Motivation

Some of the difficulties that the Art Gallery faced in their current routine are mentioned below.

- Time consuming- the business process carried out by humans. So obviously it takes more time to selections, payments settling and registrations.
- Difficult to manage stock- It is difficult to find out that what materials are end, what materials are mostly used, what kind of products are mostly sold and more.
- They manually update the payments. So it is difficult for large calculations. And lead to financial fraud.
- High amount of data redundancy occurred.
- Due to paper based process they want more storage for files.
- There was a high risk of misplacing, losing of most important information.
- Difficulty in generating comprehensive reports because of manual process it takes long time to generate reports and, due to that the upper management faced many problems in making decisions.
- Searching/finding for particular product, artist, supplier or other information is a nightmare.

5. Objectives

The solutions that are provided to the above-mentioned problems through my system are listed below.

- Registration can be done easily and it just 2 minutes.
- Reduce data redundancy and errors.

- Improve the coordination between purchasing, production, sales and finance.
- Provide accurate and up-to-date information to management.
- Details can be found within 2 minutes.
- Security is high due to the access of sensitive and important data is only for authorized people.
- Reduce paper cost by 98% because all the details are stored in the database, not in the files.
- Automate and generate meaningful business report for analysis.
- Reduce 60% of manual workload.

6. Scope

The following list outlines the scopes covered by the proposed system.

- User logging management
- Production management
- Supplier management
- Supplier payment management
- Product purchase management
- Inventory management
- Quotation management
- Manage artists/customer details
- Product details management
- Manage customer payments
- Manage artist payments
- Sales management/ Invoice management
- Generate reports
- Employee management
- Privilege management

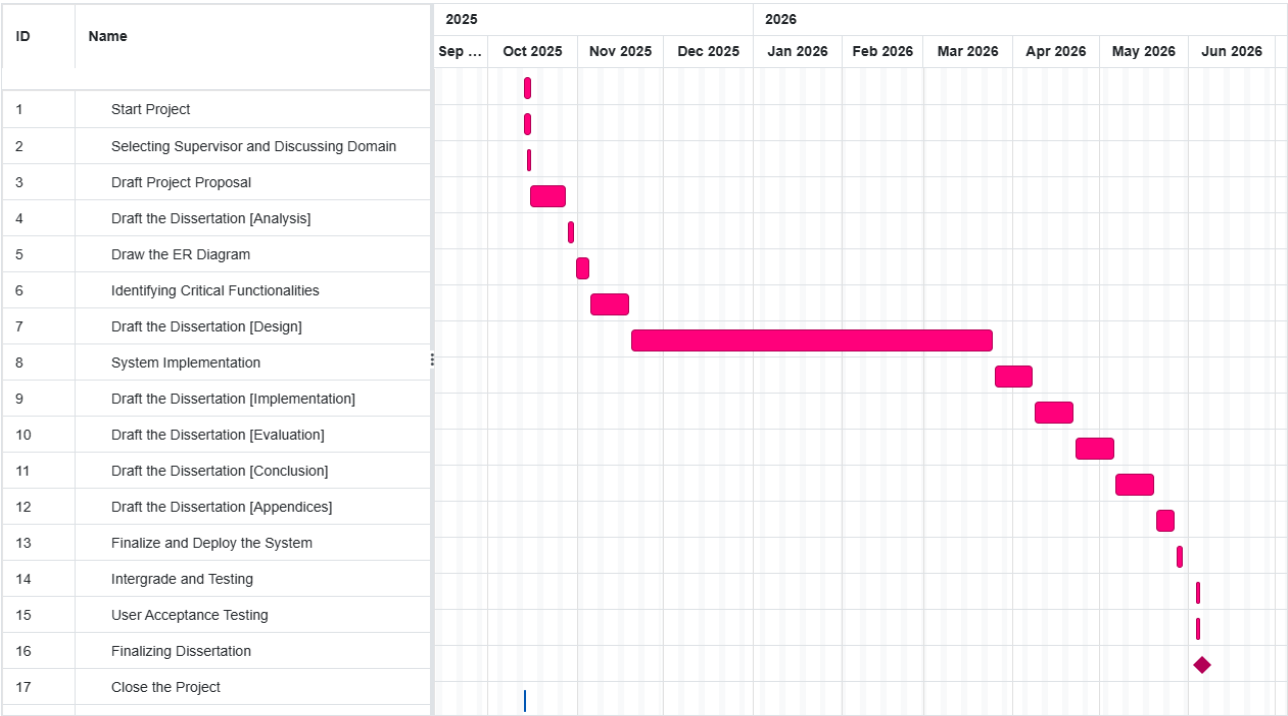
7. Critical Functionalities

- Supplier management
- Production management
- Payment management
- Purchase management
- Report generation
- Manage registration
- Inventory management
- Item management/Invoice management

8. Deliverables

- Full functional system
- System documentation
- Staff and volunteer training
- Reference manuals
- Software tools to help maintenance of the code

9. Gantt Chart



10. Resource Requirements

Software

- MySQL Work Bench 8.0
- MySQL 8.0
- Windows 10 Pro Operating System
- Intel J Idea
- Microsoft Project 2019
- Visual Paradigm
- Gradle (Dependency Manager)
- Java 21
- Google Chrome / Microsoft Edge
- Spring Framework
- Bootstrap 5

Hardware

- Computer with processing power similar or higher than 2.0 GHz
- RAM with 4GB or above
- Processor with core i3
- Hard Disk 1TB or above
- Graphics Card – Intel(R) HD Graphics 5500 (128MB)

11. Evaluation Criteria

Test Plan

- Test scenarios will be created and executed to validate system functionalities.
- Unit testing will be done while during the development of individual modules.
- Integration testing will be conducted after developing modules.
- System testing will be done after completing the system.
- Acceptance testing will be carried out before delivering the system.

Supervisor meeting plan

- By analyzing the business to have a better understanding of the scope of the project.
- Identify the critical functionalities of the proposed system.
- Discuss tools, frameworks and techniques used.
- Address skills and knowledge required for development.